

Chapter 2 Homework Assignment

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Problem 1: Transformations and Standardization

Let X be a continuous random variable.

1. Prove $Var(aX + b) = a^2 Var(X)$.

Sol.

$$\begin{aligned} Var(aX + b) &= E[(aX + b)^2] - E[(aX + b)]^2 \\ &= E[a^2 X^2 + 2abX + b^2] - [aE(X) + b]^2 \\ &= a^2 E[X^2] + 2abE[X] + b^2 - (a^2 E[X]^2 + 2abE[X] + b^2) \\ &= a^2 (E[X^2] - E[X]^2) \\ &= a^2 Var(X) \end{aligned}$$

2. Let $Y = \frac{X-\mu}{\sigma}$. Show that $E[Y] = 0$ and $Var(Y) = 1$.

Sol.

Let $\mu = E[X]$, $\sigma = \sqrt{Var(X)} > 0$, therefore

$$E[Y] = E\left[\frac{X-\mu}{\sigma}\right] = \frac{1}{\sigma}(E[X] - \mu) = \frac{1}{\sigma}(\mu - \mu) = 0,$$

$$Var(Y) = Var\left(\frac{1}{\sigma}X - \frac{\mu}{\sigma}\right) = \left(\frac{1}{\sigma}\right)^2 Var(X - \mu) = \frac{1}{\sigma^2} Var(X) = \frac{1}{\sigma^2} \sigma^2 = 1.$$

3. If $X \sim \text{Exp}(\lambda)$ derive the CDF and verify the memoryless property.

Sol.

CDF:

$$F_X(x) = P(X \leq x) = \int_0^x \lambda e^{-\lambda t} dt = [-e^{-\lambda t}]_0^x = 1 - e^{-\lambda x}.$$

Thus

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-\lambda x}, & x \geq 0 \end{cases}.$$

Let $s, t \geq 0$. Then

$$P(X > s + t \mid X > s) = \frac{P(X > s + t)}{P(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t).$$

Hence the exponential distribution is memoryless.

SPC Link: Why is standardization fundamental in control limits?

Sol.

在 SPC 中，標準化重要的原因是：控制界線是建立在機率的基礎上，透過 $\frac{X-\mu}{\sigma}$ 的轉換，我們可以把統計量的中心移至 0，並以標準差為距離作為統一的單位，使不同製程資料都能在相同的標準下判斷是否異常，這也是為甚麼常見的 $3\text{-}\sigma$ 控制界線具有意義。

Problem 2: Normal Sampling Theory

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$

1. Derive the distribution of $\bar{X}$.

Sol.

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n} = \frac{X_1 + X_2 + \dots + X_n}{n}, \text{ then}$$

$$E[\bar{X}] = \frac{E[X_1 + X_2 + \dots + X_n]}{n} = \frac{n\mu}{n} = \mu,$$

$$\text{Var}(\bar{X}) = \frac{\text{Var}[X_1 + X_2 + \dots + X_n]}{n^2} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

Hence

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

2. Show $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$.

Sol.

Let $S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$, and we already know the following:

(1) $\bar{X}$ and S^2 are independent by Basu's theorem.

- (2) If $Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$, then $Z^2 \sim \chi_1^2$.
(3) If $Y_i \stackrel{iid}{\sim} \chi_1^2$, then $\sum_{i=1}^n Y_i \sim \chi_n^2$, $i = 1, \dots, n$.
(4) If $Q \sim \chi_k^2$, then $M_Q(t) = (1 - 2t)^{-k/2}$ for $t < \frac{1}{2}$.

Compute

$$\begin{aligned} \sum_{i=1}^n (X_i - \mu)^2 &= \sum_{i=1}^n ((X_i - \bar{X}) + (\bar{X} - \mu))^2 \\ &= \sum_{i=1}^n (X_i - \bar{X})^2 + n(\bar{X} - \mu)^2. \end{aligned}$$

Then divide by σ^2 :

$$\sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 = \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma} \right)^2 + \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right)^2.$$

Let

$$U = \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2, \quad V = \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma} \right)^2, \quad W = \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right)^2.$$

Then $U = V + W$, and we can know

$$U = \sum_{i=1}^n Z_i^2 \sim \chi_n^2, \quad V = \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{(n-1)S^2}{\sigma^2},$$

also $\bar{X} \sim N(\mu, \sigma^2/n)$, hence

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1) \Rightarrow W \sim \chi_1^2.$$

By (1) and W is a function of $\bar{X}$ while $V = \frac{(n-1)S^2}{\sigma^2}$ is a function of S^2 , we have

$$V \perp\!\!\!\perp W.$$

From $U = V + W$ and independence, we have

$$M_U(t) = M_V(t) M_W(t).$$

By (4), we have

$$(1 - 2t)^{-n/2} = M_U(t) = M_V(t) (1 - 2t)^{-1/2} \Rightarrow M_V(t) = (1 - 2t)^{-(n-1)/2}.$$

Since $M_V(t)$ is the mgf of χ_{n-1}^2 , finally,

$$V = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2.$$

3. Deduce $T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$.

Sol.

Let $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$. From Q2, we know and let $U = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$.

Compute

$$\begin{aligned} T = \frac{\bar{X} - \mu}{S/\sqrt{n}} &= \frac{(\bar{X} - \mu)/(\sigma/\sqrt{n})}{(S/\sqrt{n})/(\sigma/\sqrt{n})} = \frac{Z}{S/\sigma} = \frac{Z}{\sqrt{(S/\sigma)^2}} \\ &= \frac{Z}{\sqrt{(n-1)S^2/(n-1)\sigma^2}} \\ &= \frac{Z}{\sqrt{U/(n-1)}} \end{aligned}$$

Because of the definition of t -distribution : if $Z \sim N(0, 1)$ 、 $U \sim \chi_{\nu}^2$ and $Z \perp U$, then

$$\frac{Z}{\sqrt{U/\nu}} \sim t_{\nu}.$$

In conclusion,

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{Z}{\sqrt{U/(n-1)}} \sim t_{n-1}$$

SPC Link: Known vs unknown σ changes limits.

Sol.

在 SPC 中，若製程資料的標準差 σ 已知，則 $\bar{X}$ 的控制界線會服從常態分配 Z ；若 σ 未知，則會以樣本標準差 S 取代 σ 來估計，因此統計量會服從 t 分配，因此控制界線需要調整。

Problem 3: Discrete Structure and Limits

1. Derive $E[X]$, $Var(X)$ for $X \sim \text{Bin}(n, p)$.

Sol.

Let $K_i \stackrel{iid}{\sim} \text{Bernoulli}(p)$, we compute

$$E[K_i] = 1 \times p + 0 \times (1 - p) = p,$$

$$E[K_i]^2 = 1^2 \times p + 0^2 \times (1 - p) = p,$$

$$Var(K_i) = p - p^2 = p(1 - p).$$

Since $X = \sum_{i=1}^n K_i$. Therefore,

$$E[X] = E\left[\sum_{i=1}^n K_i\right] = \sum_{i=1}^n E[K_i] = \sum_{i=1}^n p = np,$$

$$Var(X) = Var\left(\sum_{i=1}^n K_i\right) = \sum_{i=1}^n Var(K_i) = \sum_{i=1}^n p(1-p) = np(1-p).$$

2. Prove the Poisson limit of Binomial.

Sol.

Let $X_n \sim \text{Bin}(n, p)$ with $np \rightarrow \lambda > 0$ as $n \rightarrow \infty$ and $p \rightarrow 0$.

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

We compute

$$\binom{n}{k} p^k = \frac{n(n-1)\cdots(n-k+1)}{k!} p^k = \frac{1}{k!} \left(\prod_{j=0}^{k-1} \frac{n-j}{n} \right) (np)^k \rightarrow \frac{1}{k!} \lambda^k, \text{ as } n \rightarrow \infty,$$

$$(1-p)^{n-k} = (1-p)^n (1-p)^{-k} = \left[(1-p)^{1/p} \right]^{np} (1-p)^{-k} \rightarrow e^{-\lambda} \cdot 1, \text{ as } n \rightarrow \infty.$$

In conclusion,

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k} \rightarrow \frac{e^{-\lambda} \lambda^k}{k!}, \text{ as } n \rightarrow \infty.$$

3. Give an SPC context favoring Poisson modeling.

Sol.

在統計製程管制當中，Poisson 分配最常被用來描述在固定時間或固定區域內，某事件發生的次數

例如，在半導體製程產線中，工程師檢查每片晶片，並記錄晶片表面上的缺陷數量。

令 X 為每片晶片的缺陷數量，若缺陷在晶片表上隨機且彼此間獨立，且在固定區域內的缺陷發生數近似於常數，令 λ 為每片晶片的平均缺陷數，則

$$X \sim \text{Poisson}(\lambda) \Rightarrow P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

Problem 4: Mixtures and Overdispersion

1. Show Poisson–Gamma $\Rightarrow$ Negative Binomial.

Sol.

Let $X \mid \Lambda = \lambda \sim \text{Pois}(\lambda)$ and $\Lambda \sim \text{Gamma}(\alpha, \beta)$ for $\alpha, \beta > 0$, i.e.

$$f_{\Lambda}(\lambda) = \frac{e^{-\beta\lambda}\beta^{\alpha}}{\Gamma(\alpha)}\lambda^{\alpha-1}, \quad \lambda > 0.$$

Then

$$f_{X|\Lambda}(x \mid \lambda) = \frac{e^{-\lambda}\lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

So we can compute

$$\begin{aligned} P(X = x) &= \int_0^{\infty} f_{X|\Lambda}(x \mid \lambda) f_{\Lambda}(\lambda) d\lambda \\ &= \int_0^{\infty} \frac{e^{-\lambda}\lambda^x}{x!} \cdot \frac{e^{-\beta\lambda}\beta^{\alpha}}{\Gamma(\alpha)} \lambda^{\alpha-1} d\lambda \\ &= \frac{\beta^{\alpha}}{x! \Gamma(\alpha)} \int_0^{\infty} \lambda^{x+\alpha-1} e^{-(\beta+1)\lambda} d\lambda. \end{aligned}$$

Use $\int_0^{\infty} \lambda^{t-1} e^{-k\lambda} d\lambda = \frac{\Gamma(t)}{k^t}$ with $t = x + \alpha$ and $k = \beta + 1$:

$$\begin{aligned} P(X = x) &= \frac{\beta^{\alpha}}{x! \Gamma(\alpha)} \cdot \frac{\Gamma(x + \alpha)}{(\beta + 1)^{x+\alpha}} \\ &= \frac{\Gamma(x + \alpha)}{\Gamma(\alpha) x!} \cdot \left(\frac{\beta}{\beta + 1} \right)^{\alpha} \left(\frac{1}{\beta + 1} \right)^x. \end{aligned}$$

Let

$$p = \frac{\beta}{\beta + 1}, \quad 1 - p = \frac{1}{\beta + 1}, \quad p \in [0, 1].$$

Therefore

$$P(X = x) = \frac{\Gamma(x + \alpha)}{\Gamma(\alpha) x!} p^{\alpha} (1 - p)^x, \quad x = 0, 1, 2, \dots$$

It is the pmf of a Negative Binomial distribution:

$X \sim \text{NegBin}(\alpha, p)$ for number of failures x before α successes

2. Show Beta–Binomial via marginalization.

Sol.

Let

$$X \mid P = p \sim \text{Bin}(n, p), \quad P \sim \text{Beta}(\alpha, \beta), \quad \alpha, \beta > 0,$$

where

$$f_P(p) = \frac{1}{B(\alpha, \beta)} p^{\alpha-1} (1-p)^{\beta-1}, \quad 0 < p < 1,$$

and

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}.$$

Also,

$$P_{X|P}(x \mid p) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n.$$

Then we compute

$$\begin{aligned} P(X = x) &= \int_0^1 P_{X|P}(x \mid p) f_P(p) dp \\ &= \int_0^1 \binom{n}{x} p^x (1-p)^{n-x} \cdot \frac{1}{B(\alpha, \beta)} p^{\alpha-1} (1-p)^{\beta-1} dp \\ &= \binom{n}{x} \frac{1}{B(\alpha, \beta)} \int_0^1 p^{x+\alpha-1} (1-p)^{n-x+\beta-1} dp \\ &= \binom{n}{x} \frac{B(x+\alpha, n-x+\beta)}{B(\alpha, \beta)} \\ &= \binom{n}{x} \frac{\Gamma(x+\alpha)\Gamma(n-x+\beta)}{\Gamma(n+\alpha+\beta)} \cdot \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}, \quad x = 0, 1, \dots, n. \end{aligned}$$

It is the Beta-Binomial pmf.

3. Explain why mixtures increase variance.

Sol.

混合分配會使變異數增加，乃至於 Law of total variance：

$$\text{Var}(X) = E[\text{Var}(X \mid \theta)] + \text{Var}(E[X \mid \theta]).$$

其中第一項是「抽樣變異」(Sampling variation)，描述在參數 θ 固定時的隨機變動；第二項是「異質性變異」(Heterogeneity effect)，描述在參數 θ 隨機時而引入的額外變動。

若不混合，此時 θ 為常數，則 $E[X \mid \theta]$ 為常數，因此

$$\text{Var}(E[X \mid \theta]) = 0 \Rightarrow \text{Var}(X) = E[\text{Var}(X \mid \theta)].$$

若混合，此時 θ 為隨機變數且會影響 $E[X | \theta]$ ，則

$$Var(E[X | \theta]) > 0 \Rightarrow Var(X) = E[Var(X | \theta)] + Var(E[X | \theta]) > E[Var(X | \theta)].$$

因此混合會增加異質性變異，使得總變異數膨脹（Overdispersion）。

以 Problem 4 的 Poisson-Gamma mixture 為例

已知

$$E[X | \Lambda] = Var(X | \Lambda) = \Lambda, \quad E[\Lambda] = \frac{\alpha}{\beta}, \quad Var(\Lambda) = \frac{\alpha}{\beta^2}$$

可得

$$\begin{aligned} Var(X) &= E[Var(X | \Lambda)] + Var(E[X | \Lambda]) \\ &= E[\Lambda] + Var(\Lambda) \\ &= \frac{\alpha}{\beta} + \frac{\alpha}{\beta^2} \end{aligned}$$

我們知道

$$E[X] = E(E[X | \Lambda]) = E[\Lambda] = \frac{\alpha}{\beta}.$$

因此

$$Var(X) = E[X] + \frac{\alpha}{\beta^2} > E[X].$$

總結來說，對於一個 Poisson 分配，其變異數等於期望值，但經過 Poisson-Gamma 的混和分配後，造成變異數大於期望值，亦即變異數膨脹。

SPC Link: Overdispersion violates p-chart assumptions.

Sol.

在 SPC 中，c-chart 假設每個檢驗單位的缺陷數服從 Poisson 分配，其平均數與變異數皆為 λ ，若缺陷發生率在不同時間或批次之間產生變動，則會形成 Poisson-Gamma 的混和分配，使得分配變為 Negative-Binomial，且變異數會大於平均數，即 Overdispersion，因此違反 c-chart 的 Poisson 假設，使得控制界線仍在使用相對較小的變異數，導致控制界線過窄。